Assessment of Data Augmentation Paradigms in Pathology Identification

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2024 OITS International Conference on Information Technology (OCIT) | 979-8-3315-1040-4/24/$31.00 ©2024 IEEE | DOI: 10.1109/OCIT65031.2024.00080

***Abstract*—Data augmentation has an important role in improving the performance of models in medical realm. This research shows the impact of various data augmentation techniques by evaluating its performance using two medical datasets. By employing augmentation strategies such as under sampling and oversampling techniques to balance imbalanced datasets which includes Repeated Edited Nearest Neighbors (RENN), Random under sampling (RUS), Near Miss and SVM, Borderline, Simple Smote techniques. This study explores the connection between data pre-processing, model development and the effectiveness of various data augmentation techniques. Three different deep learning models—Artificial Neural Networks (ANN), Long Short-Term Memory networks (LSTM), and its varient Gated Recurrent Units (LSTM) were used on each method, and their performance was observed and analyzed. The comparison of these models on different augmentation techniques provides crucial insights into how each method affects the models performance. These insights can be useful in the real time where the data is not distributed equally such as in medical, security industry. In this research two medical datasets related to heart and cancer were taken and experimented with sampling methods which have imbalance labels.**

***Index Terms*—medical realm; data augmentation; under sampling; oversampling; deep learning models**

I. Introduction

An important method in machine learning is data augmentation [1]. By introducing controlled variations, data augmentation helps models better handle real-world scenarios and limited datasets, reducing over-fitting and improving accuracy. There are many data augmentation techniques [2], in which, some of them are sampling, flipping, Gaussian noise addition, etc. The act of choosing a subset of components from a larger population in order to infer or draw conclusions about that population is referred to as sampling [3]. Sampling is often used to create representative datasets for training and testing models.

Two sampling strategies are chosen for this research, Under-sampling and Over-sampling [4]. Oversampling is an approach which focuses mainly on boosting the minority class's representation. Smote is one of the most commonly used oversampling techniques [5]. Due to its diverse applications and effectiveness in addressing class imbalance issue. Under-sampling [6] is another class imbalance preventing technique in machine learning that involves reducing the number of examples in the majority class to match the number of examples in the minority class. Some of the Under-sampling techniques used are Random Under sampling, Near Miss and Repeated Edited Nearest Neighbors (RENN). In this research explores different augmentation strategies with three different deep learning models, and compare the results of models trained with each of those layers on different tabular datasets that are gone through those different augmentation techniques [7].

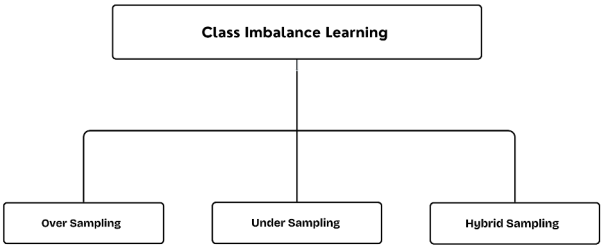

Fig. 1. Techniques in Class Imbalance Learning

The rest of the paper is organized as follows. In section II, some related models are described which gives motivation to study the challenges more. In section III, the proposed model is explained. Further, In section V, the model performance is highlighted. Finally, the paper is concluded in section VI with some future directions.

II. Literature Review

Class inequality is a serious issue in machine learning, especially in domains such as medical diagnosis, fraud detection and intrusion detection [8]. This imbalance, where one majority class significantly outweighs another minority class, can lead to bias that under perform on the minority class. To address this, many techniques have been developed to balance the label distribution and improve the model. One of the early pioneer methods is SMOTE (Chawla et al., 2002) [9], which produces synthetic data points for the minority class through interpolation between pre-existing points. This method helps to balance class distribution and improve model accuracy on least

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dominant class. Han et al. (2005) further tuned this technique with Borderline-SMOTE, which focuses on minority class data points that are near the decision boundary between classes, as these are often more crucial for better classification [10] [11]. ADASYN, introduced later, adapts to the difficulty of classification by adjusting the number of synthetic data points based on the difficulty of classifying existing minority class instances [12]. This method shows generating more synthetic points for harder to classify the instances. Mikołajczyk and Grochowski's study on deep learning based image classification demonstrated that data augmentation could improve model performance even with imbalanced datasets [1] [13]. Similarly, Liu et al. (2020) and Alomar et al. (2023) found that text data augmentation improves deep learning model performance, highlighting its effectiveness across different types of data [2] [14]. Oversampling methods, such as those discussed by Singh et al. (2022), are among the best strategies for mitigating class imbalance problems. Their survey showed that oversampling methods, including SMOTE and its variants, are crucial for balancing class distribution and enhancing model performance [3] [15]. Deep learning models, particularly in Artificial Neural Networks (ANNs) and Long Short Term Memory Networks (LSTMs), are prone to class imbalance due to their dependence on huge amounts of data, which can often be skewed sometimes. Under-sampling techniques also play a critical role in managing imbalance. Edited Nearest Neighbors (ENN) is one such method that removes data points from the majority class that are close to minority class instances, thereby preserving decision boundaries and improving classification performance (Fawad-ul-Hassan Gillani et al., 2022) [6]. Liu et al. (2020) explored model-based synthetic sampling, which has shown promise for imbalanced data classification, while Srinilta and Kanharattanachai (2021) investigated SMOTE techniques with a Natural Neighbor-based Algorithm (NNA) [16] [17] Karim et al. (2022) demonstrated the application of SMOTE in credit risk datasets, showcasing its versatility across various domains [18] [12]. M et al. (2022) combined cost-sensitive machine learning with SMOTE-SVM to address class imbalance in SCADA systems [19] [20].

III. Experimental Setup

For this research three models ANN, LSTM and GRU are used in each sampling technique. The research begins by collecting datasets from various source like Kaggle, Papers with code etc. The next step is data pre-processing which involves handling missing values, standardizing all the features, and splitting the dataset into training and test data. The next step is applying each data augmentation method on each dataset. Later, model development is done, where each model architecture is defined with particular parameters such as number of layers, activation functions like ReLU for ANN, Tanh for LSTM and GRU, and Sigmoid for all the output layers and usgae of optimizers such as Adam along with learning rates. Figure 2 shows the overall flowchart of the procedure.

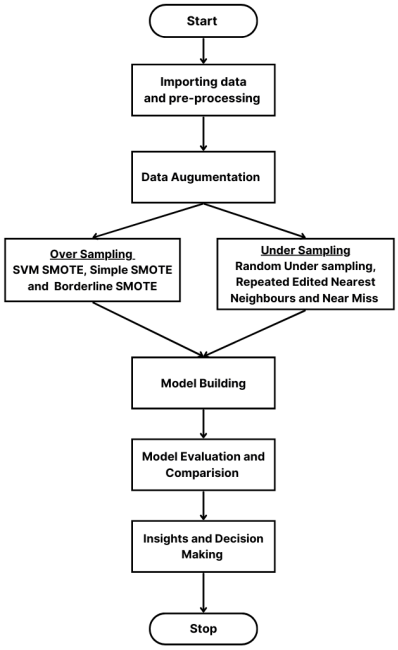

Fig. 2. Proposed Experimentation

Given below is the detailed description of each sampling method.

A. Over-Sampling

In over-sampling, SMOTE (Synthetic Minority Over-sampling Technique), a machine learning method for addressing class imbalance in datasets. When one class has significantly fewer samples than another, machine learning models may struggle to properly learn the minority class [11] [18]. Several SMOTE techniques and their combinations exist. For this research focused on the following commonly used and successful sub-techniques:

1) Simple SMOTE: Simple SMOTE is the original and basic form of the Synthetic Minority Over-sampling Technique. It works by selecting instances from the underrepresented group that are similar or close in the feature space, and generating new artificial examples at random points along the line segment connecting them. This method over samples the minority class by pushing the classifier to create larger decision regions around neighboring minority class points.

2) Borderline SMOTE: Borderline SMOTE is an extension of the original SMOTE, designed to focus on oversampling the "borderline" minority class examples that are likely to be misclassified. This technique addresses the overlap between classes, where examples near the decision boundary are more prone to miss-classification. Borderline SMOTE identifies these borderline examples using a nearest neighbors classifier (KNN) and generates synthetic examples between them and their nearest neighbors.

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3) SVM SMOTE: SVM SMOTE is a variation of Borderline SMOTE that uses a Support Vector Machine (SVM) instead of KNN to more accurately identify support vectors on the decision boundary. It generates synthetic examples between these support vectors, extending the minority class region towards the majority class. This technique extrapolates to produce additional instances if the majority class instances are less than half of their nearest neighbors.

B. Under-Sampling

For under-sampling, Random Under-sampling, Near Miss, and Repeated Edited Nearest Neighbors (RENN) algorithms have been used. Description of the techniques given below.

1) Random Under-Sampling: Random Under-sampling involves randomly selecting and removing instances from the majority class until the desired class balance is achieved, such as a 1:1 ratio between minority and majority classes. This method is fast and simple but may discard potentially useful majority class examples.

2) Near Miss: Near Miss identifies majority class examples that are closest to minority class examples in the feature space and retains them, discarding other majority class samples. This method focuses on retaining relevant majority class instances that are crucial for the classification task.

3) Repeated Edited Nearest Neighbors: The Repeated Edited Nearest Neighbors (RENN) algorithm is a modified version of the Edited Nearest Neighbors algorithm. It removes noisy, borderline, and redundant samples from both classes. For each sample, it identifies the 3 nearest neighbors—1 from the same class and 2 from the opposite class. If the nearest same-class neighbor is farther than the nearest opposite-class neighbors, the sample is removed. This process is repeated on the cleaned dataset, retaining only safe samples away from decision boundaries.

IV. Methodology

The process was started by taking each dataset individually and performing pre-processing and feature selection which are crucial procedures to make sure that the models can perform effectively. Techniques such as handling missing data through imputation (mean, median) or removing incomplete rows and standardizing numerical features to address in-differences in the scale. Categorical data were encoded using label encoding [21]. For feature selection, correlation analysis and recursive feature elimination were used to improve training speed [22]. Later, two datasets were taken and applied both sampling techniques by building three deep learning models. The description of the datasets which were used is given below.

1) Heart Stroke Prediction Dataset: The Stroke Prediction Dataset is designed to predict the likelihood of stroke occurrence based on various health and lifestyle factors. It includes 16 features and a target label column indicating whether the patient had a stroke.

TABLE I
FEATURES IN STROKE PREDICTION DATASET

Feature	Definition
id	Patient ID
age	Age of the patient
hypertension	Presence of high blood pressure (0/1)
heart_disease	Presence of heart disease (0/1)
ever_married	Marital status
work_type	Type of work
residence_type	Type of residence area
avg_glucose_level	Average glucose level
bmi	Body Mass Index
smoking_status	Smoking status
stroke	Target variable indicating if a stroke occurred (0/1)

This dataset combines medical measurements with lifestyle factors, providing a detailed view of stroke. The dataset is crucial for exploring how predictive modeling can integrate clinical and lifestyle factors to assess stroke risk. The comprehensive set of features allows researchers to evaluate the impact of various factors on stroke prediction, including medical history and lifestyle choices

2) Breast Cancer Dataset: The Breast Cancer Wisconsin Dataset focuses on the classification of breast cancer as malignant or benign based on cell nuclei measurements. It contains 32 features and one label column, which indicates the diagnosis (Benign or Malignant).

TABLE II
FEATURES IN BREAST CANCER DATASET

Feature	Definition
id	Patient ID
radius_mean	Average radius of cell nuclei
texture_mean	Average texture of cell nuclei
perimeter_mean	Average perimeter of cell nuclei
area_mean	Average area of cell nuclei
smoothness_mean	Average smoothness of cell nuclei
compactness_mean	Average compactness of cell nuclei
concavity_mean	Average concavity of cell nuclei
concave_points_mean	Average number of concave points of cell nuclei
symmetry_mean	Average symmetry of cell nuclei
fractal_dimension_mean	Average fractal dimension of cell nuclei
radius_se	Standard error of radius
texture_se	Standard error of texture
perimeter_se	Standard error of perimeter
area_se	Standard error of area
smoothness_se	Standard error of smoothness
compactness_se	Standard error of compactness
concavity_se	Standard error of concavity
concave_points_se	Standard error of concave points
symmetry_se	Standard error of symmetry
fractal_dimension_se	Standard error of fractal dimension
radius_worst	Worst-case radius
texture_worst	Worst-case texture
perimeter_worst	Worst-case perimeter
area_worst	Worst-case area
smoothness_worst	Worst-case smoothness
compactness_worst	Worst-case compactness
concavity_worst	Worst-case concavity
concave_points_worst	Worst-case concave points
symmetry_worst	Worst-case symmetry
fractal_dimension_worst	Worst-case fractal dimension
diagnosis	Target variable indicating diagnosis (B= Benign, M= Malignant)

This dataset is pivotal for evaluating how different data augmentation techniques affect the performance of deep learning models in medical diagnosis, particularly for classifying Benign and Malignant.

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TABLE III
CLASS DISTRIBUTION IN STROKE AND BREAST CANCER DATASETS

Dataset	Output Label	Number of Samples
Stroke Dataset	No Stroke	4861
	Stroke	249
Breast Cancer Dataset	Benign	357
	Malignant	212

In the Stroke Dataset, in total there are 4861 samples of ”No Stroke” and 249 samples of ”Stroke”. In case of Breast Cancer Dataset, 357 instances are ”Benign” and 212 cases of ”Malignant” samples.

V. RESULTS AND ANALYSIS

Experiments were performed on each model using each method in over-sampling and under-sampling. The results were shown below.

TABLE IV
PERFORMANCE COMPARISON WITH DIFFERENT OVERSAMPLING TECHNIQUES ON STROKE PREDICTION DATASET

Oversampling Method	Model	Precision	Recall	F1 Score	Accuracy
SVM	ANN	90.85	91.82	90.83	87.8
	LSTM	88.77	86.80	87.79	84.1
	GRU	90.79	87.83	88.81	85.50
Simple	ANN	87.81	80.89	83.85	84.10
	LSTM	88.74	67.91	76.81	79.10
	GRU	90.73	66.92	76.82	79.10
Borderline	ANN	93.87	86.94	89.90	89.97
	LSTM	88.83	82.89	84.86	85.00
	GRU	87.86	86.87	87.87	86.70

TABLE V
PERFORMANCE COMPARISON WITH DIFFERENT UNDERSAMPLING TECHNIQUES ON STROKE PREDICTION DATASET

Undersampling Method	Model	Precision	Recall	F1 Score	Accuracy
RUS	ANN	83.68	66.85	73.76	74.6
	LSTM	83.65	58.90	70.76	73.0
	GRU	87.65	58.90	70.76	73.0
RENN	ANN	96.89	95.23	98.37	95.6
	LSTM	95.75	94.4	97.8	94.60
	GRU	95.00	95.1	97.8	94.6
NM	ANN	83.94	96.78	89.85	87.3
	LSTM	80.77	79.78	80.77	78.5
	GRU	80.96	97.73	88.83	85.7

In the Stroke Prediction dataset, the oversampling techniques present interesting results. Borderline SMOTE remains one of the top-performing techniques. The SVM- based approach in SMOTE contributes to better identifying informative samples for generating synthetic instances, resulting in improved overall model performance. However, SVM SMOTE stands out as a competitive alternative. While SVM SMOTE maintains its superiority in overall accuracy, Better results in certain metrics pertaining to finding positive instances (recall) and striking a balance between precision and recall are displayed by borderline SMOTE (F1 score). The ”borderline” minority class instances that are most likely to be incorrectly identified are oversampled in the Borderline SMOTE approach. In the context of predicting strokes, where identifying positive cases is crucial, Borderline SMOTE’s emphasis on the borderline instances proves beneficial. The Stroke Prediction Dataset’s particular features, with its clinical health histories and the importance of correctly identifying positive cases, align well with the strategy employed by Borderline SMOTE. It does exceptionally well in resolving the issues brought about by the dataset’s imbalance. contributing to its competitive performance.

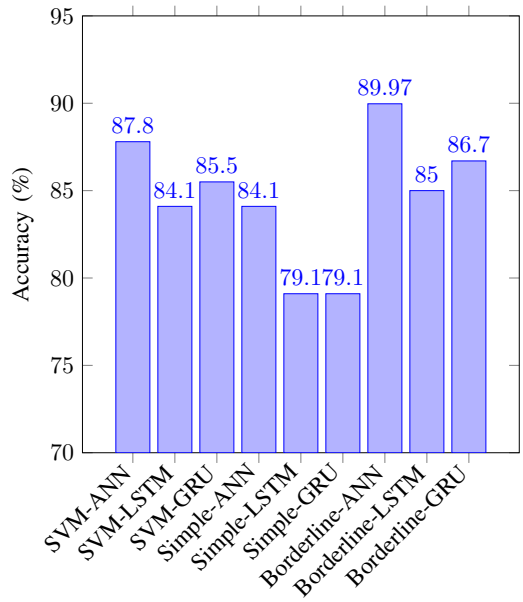

Fig. 3. Accuracy Comparison with Different Oversampling Techniques on Stroke dataset

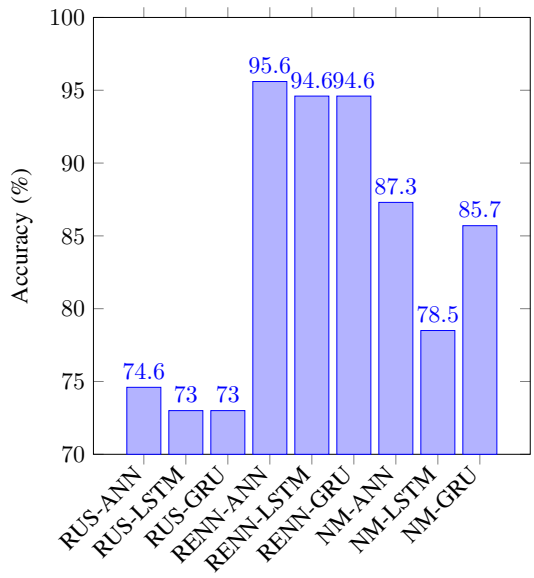

Fig. 4. Accuracy Comparison with Different Undersampling Techniques on Stroke dataset

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While SVM SMOTE remains the overall top performer, the noteworthy performance of Borderline SMOTE in certain metrics highlights the importance of selecting oversampling strategies according to the particular objectives and features of the dataset. Regarding the prognosis of stroke, where positive instances (strokes) are critical, the nuanced approach of Borderline SMOTE proves advantageous. In the Stroke Prediction Dataset's under-sampling results, Repeated Edited Nearest Neighbour (RENN) stands out as the most effective technique, surpassing Random Under Sampler and Near Miss in key metrics. A modified version of the Edited Nearest Neighbours algorithm is used by RENN, which eliminates noisy, borderline, and redundant samples from both classes. This refined approach enhances the dataset's quality by focusing on relevant samples and mitigating the impact of outliers, resulting in improved model performance. Random Under Sampler (RUS) and Near Miss exhibit competitive performance, across all deep learning models, RUS marginally outperforms Near Miss in every aspect. By lowering the proportion of instances of the majority class, both strategies seek to alleviate class imbalance and provide a more balanced training set. However, RENN's enhanced capability in removing noisy and redundant samples contributes to its superior performance in this context. Their capability to capture intricate patterns aligns well with the refined datasets created by under-sampling techniques, showcasing their robust performance in the context of stroke prediction.

TABLE VI
PERFORMANCE COMPARISON WITH DIFFERENT OVERSAMPLING TECHNIQUES ON BREAST CANCER DATASET

Oversampling Method	Model	Precision	Recall	F1 Score	Accuracy
SVM	ANN	97.98	98.97	98.98	97.6
	LSTM	82.86	87.81	85.84	84.10
	GRU	90.96	96.90	93.92	93.0
Simple	ANN	96.10	95.50	98.90	97.60
	LSTM	90.95	95.89	92.92	92.00
	GRU	95.94	94.90	94.90	94.0
Borderline	ANN	96.98	98.96	97.97	97.2
	LSTM	83.88	89.81	86.84	85.1
	GRU	97.90	97.90	98.90	98.1

TABLE VII
PERFORMANCE COMPARISON WITH DIFFERENT UNDERSAMPLING TECHNIQUES ON BREAST CANCER DATASET

Undersampling Method	Model	Precision	Recall	F1 Score	Accuracy
RUS	ANN	93.96	97.91	95.94	94.40
	LSTM	89.91	93.86	91.88	89.97
	GRU	91.87	88.90	90.88	88.90
RENN	ANN	97.10	97.96	98.90	98.10
	LSTM	92.97	98.89	95.93	91.90
	GRU	92.97	92.95	98.98	93.70
NM	ANN	91.80	99.98	94.93	93.70
	LSTM	91.90	93.91	92.91	91.30
	GRU	93.91	93.91	93.92	92.2

From the above, the oversampling techniques exhibit compelling results, with Simple SMOTE outperforming other SMOTE techniques . The success of Simple SMOTE can be attributed to its original synthetic oversampling technique, producing artificial examples between minority class cases to effectively solve class imbalance. This method performs better when it comes to diagnosing breast cancer because of its efficiency and simplicity. SVM SMOTE, although achieving competitive results, takes the second place. The SVM-based approach used in selecting support vectors on the decision boundary provides an effective strategy for generating synthetic instances, contributing to improved model performance. However, the slightly more complex nature of SVM SMOTE compared to Simple SMOTE might introduce some additional computational overhead, affecting its overall efficiency in this specific dataset. The hierarchical feature extraction and spatial relationships leveraged by different models make them suitable for extracting relevant features from this dataset, contributing to their performance.

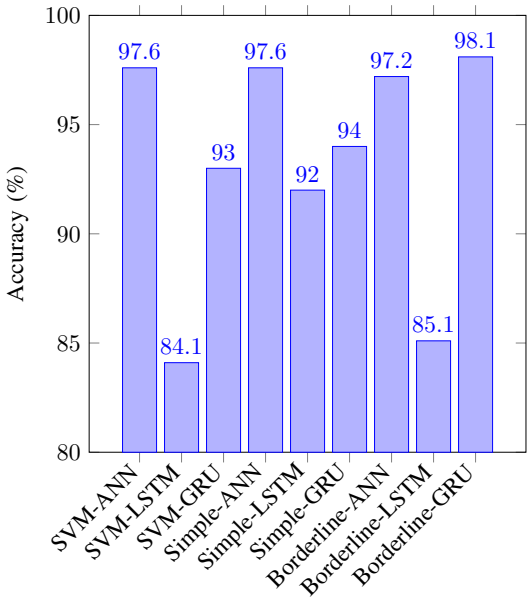

Fig. 5. Accuracy Comparison with Different Oversampling Techniques on Breast Cancer Dataset

The under-sampling techniques reveal distinctive results. Repeated Edited Nearest Neighbor (RENN) is the most effective technique. By eliminating redundant and noisy samples, the Edited Nearest Neighbors method in RENN is improved, which helps to raise the dataset's quality. Because of this improved method, RENN is the best option for under-sampling when it comes to breast cancer diagnosis. Among the under-sampling techniques, Near Miss demonstrates competitive performance but falls slightly behind RENN in terms of overall metrics. The Breast Cancer Wisconsin Dataset's features may not precisely match Near Miss's method of deleting instances of the dominant class that most closely resemble examples of the non-dominant class. This method may result in the elimination of significant majority class samples, which would impact the model's overall performance. Random Under Sampler shows competitive performance, but it does not surpass RENN in terms of overall metrics. The random removal of samples

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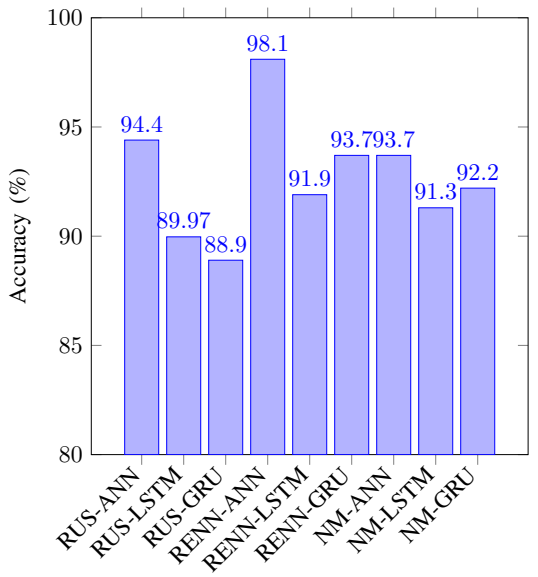

Fig. 6. Accuracy Comparison with Different Undersampling Techniques on Breast Cancer Dataset

might not be as strategic in the context of breast cancer diagnosis, where each sample holds valuable information for classification tasks.

VI. Conclusion and Future Work

In conclusion, experiments on oversampling and under-sampling methods applied to deep learning models reveals best way for organising imbalanced classification tasks. Among the oversampling methods, SVM SMOTE proves to be a good choice, offering a trade-off between recall and precision by using support vector machines in synthetic sample generation. On the under-sampling side, Repeated Edited Nearest Neighbors consistently provides high precision, making it ideal for reducing false positives while maintaining vital information. These findings highlight the importance of using both oversampling and under-sampling techniques to the specific context of the classification task. In future , hyperparameter tuning, and consideration of dataset characteristics will be done for optimizing performance in imbalanced models.

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